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The Many Clocks Interpretation of Quantum Mechanics

The Dirac Equation as a Kuramoto Phase-Synchronization System: Mass as Chiral Coupling, Measurement as Re-Synchronization

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Abstract

We demonstrate that the Dirac equation, written in the chiral (Weyl) basis, has the mathematical structure of a Kuramoto phase-synchronization system. The left- and right-handed chiral sectors act as coupled oscillators, with the mass term playing the role of the Kuramoto coupling constant: $K = m$. For massless particles ($K = 0$) the chiral clocks are permanently decoupled; for massive particles ($K > 0$) they synchronize, and the synchronized state is what we identify as a particle with definite rest mass. The Higgs-Yukawa interaction is the mechanism that sets K : $K = y_f \cdot \langle \varphi \rangle = m$.

This identification has several consequences. Quantum measurement is re-interpreted as Kuramoto re-synchronization: a particle initially synchronized with its entangled partner re-synchronizes to the macroscopic detector bulk, whose overwhelmingly greater Kuramoto inertia ($M_{\text{detector}} \gg m_{\text{particle}}$) forces conformity. Decoherence is not merely loss of phase coherence with the partner — it is the positive process of gaining coherence with the bulk. The Born rule $P = |\psi|^2$ is reframed as the long-run frequency of energy partition: $|\psi|^2$ is the energy density of the real ψ field, and the apparent stochasticity of measurement outcomes arises from unbiased zero-point + thermal background fluctuations at the synchronization event, rather than from an independent probability axiom. The Heisenberg uncertainty principle appears as a bandwidth limitation of the particle’s internal phase clock.

We present this framework — which we call the **Many Clocks Interpretation of Quantum Mechanics (MCI)** — as an interpretive reframing of standard quantum mechanics, not a modification of its predictions. The name is a deliberate echo of, and contrast to, Many-Worlds: where MWI preserves unitarity by branching the universe at each measurement, MCI preserves a single world by treating each particle as carrying its own physical phase clock, with measurement as local synchronization between clocks. No conscious observer is required; no branching occurs. The Dirac equation is unchanged; Bell’s theorem is not challenged. What is new is the identification of mass as synchronization coupling, measurement as re-synchronization dynamics, and decoherence as a Kuramoto process with a specific physical mechanism.

1. Introduction

1.1 The Measurement Problem

The quantum measurement problem remains the most consequential unsolved problem in the foundations of physics. The Schrödinger equation is linear and deterministic, yet measurements have definite outcomes. A system in superposition $\psi = \alpha|0\rangle + \beta|1\rangle$, entangled with a detector, produces

$$|\Psi\rangle = \alpha|0\rangle|D_0\rangle + \beta|1\rangle|D_1\rangle$$

The equation has no mechanism to select one outcome. The Born rule — $P(0) = |\alpha|^2$ — is postulated, not derived.

The three mainstream responses each carry significant costs. Copenhagen introduces a mysterious collapse with no physical mechanism. Everett’s many-worlds [16] preserves unitarity at the cost of infinite unobservable branches. Bohmian mechanics [15] accepts explicit nonlocality through a physically inert pilot wave.

1.2 What Is Missing: A Dynamical Mechanism for Decoherence

Environmental decoherence [4] has become the standard framework for understanding how quantum superpositions become effectively classical. It successfully explains the emergence of preferred “pointer states” and the suppression of off-diagonal density matrix elements through interaction with environmental degrees of freedom.

Yet decoherence theory describes *what happens* without specifying the dynamical mechanism by which it happens. The environment is treated as an abstract bath with many degrees of freedom. This paper proposes that the Kuramoto phase-synchronization model — specifically, as it appears in the structure of the Dirac equation itself — provides this missing mechanism.

1.3 The Proposal in Brief

We make three claims, in decreasing order of mathematical rigor:

1. **The Dirac mass term is a Kuramoto coupling** (Section 2). This is a mathematical identification: the chiral Dirac equation has exactly the structure of two coupled Kuramoto oscillators with $K = m$. This is the strongest claim and is verifiable by inspection.
2. **Measurement is re-synchronization** (Section 3). When a particle interacts with a macroscopic detector, the Kuramoto dynamics drive the particle’s phase from coherence with its entangled partner to coherence with the detector bulk. This provides a physical picture of decoherence. This is an interpretive claim.
3. **The Born rule is reframed as energy partition** (Section 4). Reading $|\psi|^2$ as wave energy density and the apparent stochasticity as unbiased background-field fluctuation at the synchronization event, $P = |\alpha|^2$ appears as the long-run frequency of channel-energy partition rather than as an independent probability axiom. This is an interpretive claim with physical motivation, not a complete derivation.

1.4 What This Paper Does Not Claim

We state clearly at the outset:

- We do not challenge Bell’s theorem [9]. The quantum correlations $E(a,b) = -\cos(a-b)$ arise from the entangled singlet state and the full Dirac spinor structure — standard quantum mechanics.
- We do not modify the Dirac equation or any of its predictions.
- We do not propose new physics. We propose a new *mechanism* for known physics.
- The framework is an interpretation — it may or may not generate distinguishable predictions from standard decoherence theory. We are honest about this throughout.

1.5 Relational Structure: The Many Clocks Interpretation

We name this framework the **Many Clocks Interpretation of Quantum Mechanics (MCI)**. The name is a deliberate echo of Many-Worlds, with a sharp contrast: Many Worlds preserves unitarity by branching the universe at each measurement and accepting an exponentially proliferating multiverse; the Many Clocks Interpretation preserves a single world by treating each particle as carrying its own internal phase clock, with measurement as local synchronization between clocks. The interpretation is *relational* — there is no global preferred reference frame and no observer required — and it operates by the following structural commitments:

1. **Every particle carries one or more intrinsic phase clocks.** For massive Dirac fermions, two chiral clocks (L and R), coupled by $K = m$ on the synchronized manifold (Section 2). For photons, polarization clocks coupled to detectors via $K_\gamma \sim \omega$ (Section 5). The clocks are physical oscillators, not bookkeeping devices.
2. **There is no global preferred frame.** Each interaction happens in the local frame jointly defined by the interacting parties. Each particle’s clocks are intrinsic — defined in its own rest frame — and a particle’s clock structure is described differently in different observer frames, but the physics of each interaction is invariant.
3. **Synchronization is a local, objective, physical event.** When two clock-carrying systems interact, their clocks synchronize via Kuramoto coupling at the interaction event. The synchronization is objective: distant observers in other frames may disagree about the ordering of spacelike-separated events elsewhere, but the two parties to a specific interaction always agree on what happened locally between them. This is ordinary special-relativistic causality, not novel — but worth stating because it removes the appearance of observer-dependence.
4. **No conscious observer is required. No branching is required.** A single objective physical world contains many clocks; interactions synchronize them locally; measurement is a special case of synchronization in which one party is a macroscopic detector with overwhelming Kuramoto inertia (Section 3).
5. **Past phase history is overwritten at each sync event.** After a small number of interactions, a particle’s pre-interaction phase is effectively unrecoverable. This is the framework’s specific dynamical mechanism for decoherence and thermalization: not just loss of off-diagonal coherence, but active phase-overwriting at each interaction event.
6. **All physically meaningful quantities are gauge- and Lorentz-invariant.** The individual chiral phases φ_L, φ_R shift uniformly under $U(1)$ gauge transformation; only the gauge-invariant difference $\varphi_L - \varphi_R$ enters the Kuramoto sine coupling. The L/R block decomposition of the spin operator is frame-relative under Lorentz boosts, but the sum $E_{LL} + E_{SS} + E_{LS} = -\cos(a-b)$ is Lorentz-invariant. The framework’s *predictions* live in invariant content; the interpretive language about “which clock dominates” is a frame-relative description of an invariant structure.

The Many Clocks Interpretation has affinities with Hestenes’ Zitterbewegung interpretation (Section 7.7) — both treat the complex phase factor in the Dirac wave function as a physical clock — and with Rovelli’s relational quantum mechanics [24], in the sense that no global preferred frame or observer is needed. It differs from both in its explicit dynamical mechanism (Kuramoto re-synchronization), and from Rovelli’s

RQM in keeping ψ ontic rather than relational: the wavefunction is a real oscillating field, not a relational state.

2. The Dirac Equation as a Kuramoto System

2.1 Chiral Decomposition

In the Weyl (chiral) basis, the Dirac spinor decomposes into two two-component Weyl spinors:

$$\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}$$

The Dirac equation separates into two coupled equations:

$$i\bar{\sigma}^\mu \partial_\mu \psi_L = m\psi_R \quad (1a)$$

$$i\sigma^\mu \partial_\mu \psi_R = m\psi_L \quad (1b)$$

where $\hat{\sigma}^\mu = (I, \sigma)$ and $\hat{\bar{\sigma}}^\mu = (I, -\sigma)$ are the Weyl matrices.

For $m = 0$ (massless): Equations (1a) and (1b) decouple entirely. ψ_L and ψ_R satisfy independent equations. The particle has definite chirality and no rest frame. Photons and massless-limit neutrinos are in this category.

For $m \neq 0$ (massive): The equations couple ψ_L to ψ_R and vice versa. Neither chirality state is stable alone; the physical particle is a superposition of both. The coupling strength is exactly m .

2.2 The Kuramoto Identification

The standard (classical) Kuramoto model [2] describes the phase dynamics of N coupled oscillators:

$$\frac{d\phi_i}{dt} = \omega_i + \frac{K}{N} \sum_j \sin(\phi_j - \phi_i)$$

The coupling here is *nonlinear* — $\sin(\phi_j - \phi_i)$ — whereas the chiral Dirac coupling in Equations (1a, 1b) is *linear* at the operator level: $m\psi_R$ sources the equation for ψ_L and vice versa. The Kuramoto sine therefore is not present in the Dirac equation as written. It emerges, however, as the near-equilibrium phenomenology of the linear coupling under a polar (Madelung-type) decomposition. We outline the reduction explicitly so that the regime of validity is on the record.

Apply the decomposition

$$\psi_L = \rho_L^{1/2} e^{i\phi_L} \chi_L, \quad \psi_R = \rho_R^{1/2} e^{i\phi_R} \chi_R$$

where $\rho_{\{L,R\}}$ are real amplitude densities, $\phi_{\{L,R\}}$ are clock phases, and $\chi_{\{L,R\}}$ are unit two-spinors carrying the spin orientation. Substituting into Equations (1a, 1b) and separating real and imaginary parts of the time component yields a coupled system: a continuity-like equation for the amplitudes and a phase equation for the clocks. With the standard chiral phase convention (the mass coupling carrying the implicit i factor inherited from γ^0), the phase equations take the form

$$\frac{d\phi_L}{dt} = \omega_L + K \sqrt{\rho_R/\rho_L} \sin(\phi_R - \phi_L + \delta_{CP}) \quad (2a)$$

$$\frac{d\phi_R}{dt} = \omega_R + K \sqrt{\rho_L/\rho_R} \sin(\phi_L - \phi_R - \delta_{CP}) \quad (2b)$$

with the coupling constant

$$K = y_f \cdot |\langle\phi\rangle| = y_f \cdot \frac{v}{\sqrt{2}} = m \quad (3)$$

On the synchronized manifold $\rho_{\underline{L}} \approx \rho_{\underline{R}}$ — the regime relevant to a particle that has settled into a definite mass eigenstate — the amplitude prefactors reduce to unity and Equations (2a, 2b) become the Kuramoto form for two coupled oscillators. Off this manifold, the amplitude dynamics couple back through the continuity equation, and the system is properly described by the full Madelung pair with a quantum potential. The Kuramoto picture is therefore the *synchronized-manifold reduction* of the chiral Dirac dynamics, not a restatement of the Dirac equation itself.

Equation (3) is the central identification of this paper: on the synchronized manifold, the particle's mass plays the role of the Kuramoto coupling constant between its chiral phase clocks. In SI units this reads $K = mc^2/\hbar$ — the Compton frequency — making explicit that the coupling has dimensions of frequency [time⁻¹], as required by the synchronization equations (2a–2b).

This places the framework within the broader literature on **quantum synchronization** [36, 37, 38, 39], where Kuramoto-form phase dynamics have been derived from the semiclassical limit of quantum master equations for a range of coupled-oscillator systems (optomechanical arrays, cavity-QED, driven self-sustained oscillators, trapped-ion chains). The Weyl-spinor pair is the chiral-Dirac instance of this general phenomenon: the operator-level coupling is linear, but on the synchronized subspace the effective phase dynamics are Kuramoto. The distinction between *classical* Kuramoto synchronization (sine coupling fundamental at the equation-of-motion level) and the *quantum* version employed here (sine coupling emergent on the synchronized subspace from a linear underlying Hamiltonian) should be kept explicit throughout.

The Higgs vacuum expectation value [11, 12] $\langle\varphi\rangle = v/\sqrt{2} \approx 174$ GeV drives the two chiral clocks into synchronized oscillation. Before electroweak symmetry breaking ($v \rightarrow 0$), $K \rightarrow 0$ and all fermions are massless — their chiral clocks are independent. After symmetry breaking, $K = m$ and the clocks synchronize.

2.3 Two Clocks, Not One

The central physical picture is:

A massive Dirac particle carries two internal oscillators: a **temporal phase clock** oscillating at $\omega_{\underline{t}} = E/\hbar$, and a **spatial phase clock** oscillating at $\omega_{\underline{s}} = p \cdot v/\hbar$. The mass term couples them — it is the Kuramoto synchronization term that drives the two chiral sectors toward a common phase.

This interpretation does not modify the Dirac equation. It provides an ontological reading of its mathematical structure.

2.4 The Synchronized State

At Kuramoto equilibrium, the phase difference locks:

$$\phi_L - \phi_R = \delta_{CP} \quad (\text{particles}) \quad (4a)$$

$$\phi_L - \phi_R = -\delta_{CP} \quad (\text{antiparticles}) \quad (4b)$$

For CP-symmetric interactions ($\delta_{\text{CP}} = 0$), particles and antiparticles have identical phase locks — consistent with CPT symmetry. The CP-violating phase δ_{CP} shifts the equilibrium, providing a qualitative mechanism for matter-antimatter asymmetry through differential synchronization efficiency.

The synchronization rate is set by $K = m$: heavy particles synchronize their chiral clocks faster than light particles. An electron ($m_e = 0.511$ MeV) synchronizes more slowly than a top quark ($m_t = 173$ GeV) by a factor of $\sim 3.4 \times 10^5$.

2.5 Gamma Matrices as Synchronization Operators

In the Weyl basis, the mass term in the Lagrangian is:

$$\mathcal{L}_{\text{mass}} = -m\bar{\psi}\psi = -m(\bar{\psi}_L\psi_R + \bar{\psi}_R\psi_L)$$

This is off-diagonal in chirality: it connects left to right and right to left. In the Kuramoto language, it is the coupling term that drives the two chiral oscillators toward synchronization.

The matrix γ^0 swaps the two chiral sectors:

$$\gamma^0 \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} = \begin{pmatrix} \psi_R \\ \psi_L \end{pmatrix}$$

The Dirac Hamiltonian $H = \alpha \cdot p + \beta m$ is therefore the generator of synchronized phase evolution across both clock sectors.

2.6 Summary of Established vs. New Claims

Claim	Status
Dirac mass term couples $\psi_L \leftrightarrow \psi_R$	Standard QFT; textbook
$m = 0 \rightarrow$ chiral sectors decouple	Standard QFT; textbook
Higgs-Yukawa sets fermion mass via $\langle \varphi \rangle$	Standard Model; established
Negative-energy Dirac solutions = reversed temporal phase	Standard (Feynman-Stückelberg [21, 22])
Mass term reduces to Kuramoto coupling on the synchronized manifold	New interpretation (this work)
$K = m$ identification (synchronized-manifold reduction)	New (this work)

3. Measurement as Re-Synchronization

3.1 The Physical Picture

This section develops the interpretive core of the framework. We propose that quantum measurement is a specific type of Kuramoto synchronization event:

A particle initially phase-synchronized with its entangled partner re-synchronizes to the macroscopic detector bulk.

The detector is a macroscopic system with overwhelmingly greater Kuramoto inertia ($M_{\text{detector}} \gg m_{\text{particle}}$). The measurement interaction forces the particle to conform its phase to the detector's collective phase Φ_{bulk} , just as a small oscillator entrained by a massive one must adopt the massive oscillator's frequency.

3.2 Decoherence Is Half the Story

In the standard decoherence program, measurement causes the particle to lose coherence with its entangled partner. This is correct but incomplete.

In the Kuramoto picture, the particle does not merely *lose* coherence with its partner. It *gains* coherence with the detector bulk. The process has two simultaneous aspects:

1. **De-coherence** from the entangled partner (the phase lock $\varphi_A = \varphi_B$ established at creation is broken)
2. **Co-herence** with the detector bulk (a new phase lock $\varphi_A = \Phi_bulk$ is established)

Standard decoherence theory tracks only aspect (1). The Kuramoto framework makes aspect (2) explicit: the particle’s phase is not randomized — it is redirected. The final state is not “unknown phase” but “phase locked to the bulk.” This is a definite classical state, not a mixed state that merely appears classical.

3.3 The Mass Asymmetry

The detector’s macroscopic mass provides the asymmetry that ensures definite outcomes. In Kuramoto dynamics, when a small oscillator (mass m) couples to a large oscillator (mass $M \gg m$), the small oscillator always conforms to the large one. The reverse — the detector conforming to the particle — is dynamically suppressed by a factor of m/M .

This is the physical content of “wavefunction collapse”: the Kuramoto mass asymmetry between particle and detector makes the measurement outcome definite, not because of a postulated projection, but because of the dynamical inevitability of a small oscillator locking to a massive one.

3.4 Gravitational Coherence of the Bulk

What maintains the coherence of the detector bulk itself? We propose that gravity provides this role. The gravitational interaction between the $\sim 10^{26}$ atoms in a macroscopic detector provides a collective Kuramoto coupling that maintains a common phase Φ_bulk across the detector. This is analogous to the well-known classical synchronization of metronomes on a shared massive platform.

The collective rate is obtained by pair-counting the gravitational Kuramoto coupling. Consider a bulk of total mass M composed of N atoms, each of mass $m_atom = M/N$. The gravitational potential energy between any pair (i, j) at separation r_{ij} is $U_pair = G \cdot m_atom^2 / r_{ij}$, which by the framework’s general $K = E/\hbar$ identification yields a pairwise Kuramoto rate

$$K_{pair} \sim \frac{G m_{atom}^2}{\hbar r_{ij}}$$

The bulk contains $N(N - 1)/2 \approx N^2/2$ such pairs. Approximating all pair separations by a characteristic internal scale Δz , the aggregate gravitational synchronization rate is

$$\Gamma_{grav} \sim \frac{N^2}{2} \cdot \frac{G m_{atom}^2}{\hbar \Delta z} = \frac{N^2}{2} \cdot \frac{G (M/N)^2}{\hbar \Delta z} = \frac{1}{2} \frac{G M^2}{\hbar \Delta z}$$

Dropping the order-unity factor of $1/2$ to obtain the characteristic rate:

$$\boxed{\Gamma_{grav} \sim \frac{G M^2}{\hbar \Delta z}} \quad (\text{characteristic, pair-counted})$$

The M^2 scaling is therefore not a numerical coincidence but a consequence of pair counting in a many-body sync: each pair contributes $\sim K_pair$, there are $\sim N^2$ pairs, and the atomic mass $m_atom = M/N$ enters quadratically per pair, so the total aggregate rate scales as $N^2 \cdot (M/N)^2 = M^2$.

For macroscopic objects ($M \sim 1$ kg, $\Delta z \sim 1$ m), $\Gamma_{\text{grav}} \sim 6 \times 10^{23}$ rad/s — an extremely fast rate, which is precisely why bulk coherence is maintained so effectively for macroscopic masses. The gravitational synchronization coupling grows as M^2 , ensuring that macroscopic objects are locked into collective phase coherence far more strongly than microscopic particles.

The relation to Penrose’s objective-reduction rate (Section 3.5) is one of counting: Penrose’s $E_{\text{G}} = Gm^2/\Delta x$ is the gravitational self-energy of a *single* mass configuration in superposition with itself displaced; Γ_{grav} here is the aggregate Kuramoto rate over all $N^2/2$ atomic pairs in the unsuperposed bulk, larger by precisely the pair-counting factor. The two quantities answer different questions — Penrose asks how fast a superposition collapses, Γ_{grav} asks how fast the unsuperposed bulk maintains internal phase coherence — but both follow from the same $K = E/\hbar$ identification applied to gravitational pair interactions.

3.5 Connection to Penrose-Diósi

This picture has a natural connection to Penrose’s proposal [6] that gravity causes quantum state reduction. Penrose argues that a superposition of two mass configurations with different spacetime geometries is unstable, with collapse timescale $\tau \sim \hbar/E_{\text{g}}$ where E_{g} is the gravitational self-energy of the superposition.

In the Kuramoto framework, this translates to: the gravitational bulk provides a phase reference, and any quantum superposition that would require the particle to maintain coherence against the gravitational synchronization pressure will decohere on a timescale set by the gravitational coupling strength. The Penrose energy E_{g} maps onto the Kuramoto coupling rate Γ_{grav} . The two frameworks predict the same qualitative behavior — heavier objects decohere faster — from different dynamical starting points.

The structural parallel deserves emphasis: both frameworks independently identify mass as the bridge between quantum and classical regimes, but through different mechanisms — Penrose through the gravitational self-energy instability of superposed spacetime geometries, this work through Kuramoto synchronization driven by the Dirac mass term $K = m$ originating in the Higgs-Yukawa interaction. In Penrose’s picture, a sufficiently massive superposition collapses on its own because the two branches curve spacetime differently; in the Kuramoto picture, a particle transitions to classical behavior by re-synchronizing its phase to the gravitationally coherent detector bulk, driven by the mass asymmetry $M_{\text{detector}} \gg m_{\text{particle}}$. That two independent starting points — one from general relativity and quantum gravity, the other from the internal structure of the Dirac equation — converge on mass as the agent of classicalization is suggestive that both may be pointing at the same underlying physics, even if the full unification is not yet available.

3.6 Speculative Outlook: Single-World Energy Accounting

We note briefly a speculative consequence. If measurement is resonant synchronization, then the wavefunction amplitude that does not couple to the detector (the “residual” associated with the unrealized outcome) has no resonant absorber available. We conjecture that this residual thermalizes into the broadband electromagnetic vacuum as low-frequency radiation, conserving energy within a single world without requiring Everettian branching. This proposal lacks a first-principles derivation and is offered only as a direction for future investigation.

3.7 What This Section Does Not Claim

The re-synchronization picture is interpretive. It does not:

1. Derive decoherence rates quantitatively from Kuramoto parameters (this would require a full quantum treatment of many-body Kuramoto dynamics)
2. Explain *which* outcome occurs in a given individual measurement — only the probabilities (addressed in Section 4)
3. Provide a Lorentz-covariant formulation of the measurement process

4. The Born Rule as Energy Partition under Background-Fluctuation Stochasticity

4.1 The Problem and the Reframing

The Born rule — $P = |\psi|^2$ — is the bridge between the mathematical formalism and experimental outcomes. In standard quantum mechanics it is postulated, not derived. Copenhagen asserts it. Everett’s many-worlds claims to derive it from branch counting, but this remains contested [27, 28]. The measurement problem, at its core, is the question of why probability equals the squared modulus of a complex amplitude.

This framework proposes a different reading. If the wavefunction is a *real physical oscillating field* (Section 2.3), then $|\psi|^2$ is not fundamentally a probability — it is the **energy density** of that field, exactly as for any classical wave. The probabilistic appearance of measurement outcomes arises from a separate source: unpredictable interference with the zero-point + thermal background field at the moment of synchronization with the detector. The Born rule, in this reading, is the **long-run frequency of energy partition into the available detector channels under unbiased stochastic background driving**, not a probability axiom of quantum theory.

The closest precedent for this view is Nelson’s stochastic mechanics [26], which derives the Schrödinger equation from a real diffusion process in a background stochastic field. Our framework supplies a physical identity for that field — the zero-point + thermal residual of the chiral phase clocks already present in the Kuramoto/Dirac picture (Sections 9 and 11 of the equations reference) — and reads the Born rule as the resulting energy-partition statistics, rather than as an independent postulate.

4.2 $|\psi|^2$ as Wave Energy Density

For any real oscillating field, the time-averaged energy density is proportional to the squared amplitude. A complex wavefunction $\psi = \alpha|0\rangle + \beta|1\rangle$ describes a superposed oscillator with amplitude α in channel $|0\rangle$ and amplitude β in channel $|1\rangle$. In the wave-realist reading, the energy carried by each channel is

$$E_0 \propto |\alpha|^2, \quad E_1 \propto |\beta|^2 \quad (5)$$

with total wave energy $E_{\text{total}} = E_0 + E_1 = |\alpha|^2 + |\beta|^2 = 1$ in normalized units. This is classical wave physics — no Born postulate is required to write down $|\alpha|^2$ and $|\beta|^2$ as the quantities of interest. They are the energy fractions of the field in each channel.

4.3 Synchronization Deposits the Energy into One Channel

The detector channels available for synchronization are those eigenstates of the detector whose internal phase clocks can match the particle’s. The Kuramoto re-synchronization event (Section 3) transfers wave energy from the particle to the detector, exciting one channel to its registered (“clicked”) state.

A discrete click — rather than a fractional excitation of multiple channels — arises because each detector channel is itself a quantized resonator with a discrete excitation threshold (the same threshold that defines the channel as a single detector eigenstate). The synchronization event delivers either the full quantum into one channel or none, determined by which channel “wins” the competition for the available wave energy.

4.4 Stochasticity Comes from the Background Field

Why does any particular channel win in any particular event? And why do long-run frequencies match $|\alpha|^2$ rather than some other partition?

The answer rests on the zero-point + thermal background field always present at the detector (Section 9 of the equations reference). At the instant of contact, the relative phase between the particle’s clock and the detector bulk’s clock is perturbed by unpredictable transient fluctuations of this background. These fluctuations bias the synchronization event toward one channel or another in any individual instance — but, being statistically symmetric across the relevant phase space (zero-point fluctuations are symmetric

by construction; unbiased thermal noise is symmetric by the second law), they do not favor any channel systematically.

Two consequences follow:

1. **Each individual outcome is not fundamentally random.** It is determined by the specific instantaneous configuration of background fluctuations at the moment of synchronization. We cannot, as observers, track this configuration; the outcome therefore *appears* random.
2. **Long-run frequencies converge to the energy fractions.** Over many events, the unbiased background averages out, and the frequency with which each channel is chosen converges to the energy fraction the original wave carried in that channel:

$$P(|0\rangle) = |\alpha|^2, \quad P(|1\rangle) = |\beta|^2 \quad (6)$$

read here as long-run **frequencies of energy partition** rather than as an independent probability axiom.

The Born rule, on this reading, is the visible projection of a deterministic energy redistribution in which the only stochastic input is the unobserved background field. It is not a separate postulate of quantum mechanics; it is the law of energy partition under unbiased background driving, observed as frequency statistics because the background is unobserved.

4.5 Connection to Nelson's Stochastic Mechanics

Nelson [26] showed that a real particle undergoing stochastic motion in a background field with diffusion coefficient $\nu = \hbar/(2m)$ satisfies the Schrödinger equation. The framework presented here makes that background field physically explicit (the zero-point + thermal residual of the chiral phase clocks, equations reference Sections 9 and 11) and reads the Born rule as a frequentist statement about energy partition under that background's stochastic driving. Where Nelson postulates the diffusion coefficient, this framework identifies its physical origin in the zero-point phase noise $\sigma_- \varphi(0) = 1/\sqrt{2}$ rad — itself a consequence of the Kuramoto coupling $K = m$ and the chiral clock structure.

This is structurally a hidden-variable theory, in which the hidden variable is the instantaneous configuration of the background field at the detector. As such, it must satisfy Bell's theorem — and it does, because the framework does not use this story to explain Bell correlations. The correlations themselves come from the full Dirac spinor structure (standard QM, Section 2 and Appendix A); only the *measurement statistics* of definite outcomes are accounted for via the energy-partition reading. The two stories are complementary, not competing.

4.6 The Heisenberg Uncertainty Principle as the Fourier Bandwidth Theorem

In MCI's wave-realist reading (Section 4.2), the wavefunction $\psi(x)$ is a real oscillating field. Position and momentum representations are related by Fourier transform via the de Broglie relation $p = \hbar k$:

$$\varphi(p) = \frac{1}{\sqrt{2\pi\hbar}} \int \psi(x) e^{-ipx/\hbar} dx$$

The **Fourier bandwidth theorem** of harmonic analysis — a rigorous mathematical result independent of any quantum interpretation — states that for any function $f \in L^2(\mathbb{R})$ with Fourier transform $\mathfrak{f}(k)$, the standard deviations of $|f|^2$ and $|\mathfrak{f}|^2$ (treated as probability densities on x and k respectively) satisfy

$$\sigma_x \cdot \sigma_k \geq \frac{1}{2} \quad (7)$$

with equality saturated by Gaussian wavepackets. Multiplying both sides by $\hbar$:

$$\boxed{\sigma_x \cdot \sigma_p \geq \frac{\hbar}{2}} \quad (8)$$

This is the Heisenberg uncertainty relation, **factor of $\frac{1}{2}$ exact**.

The framework's contribution here is interpretive, not derivational. Standard QM postulates $\Delta x \Delta p \geq \hbar/2$ as a separate axiom (or derives it from the operator commutator $[\hat{x}, \hat{p}] = i\hbar$, which is itself a postulate, via the Robertson–Schrödinger inequality). MCI reads it as the Fourier bandwidth theorem applied to a real wavefunction — a theorem about waves, not an axiom about quantum systems. **In the wave-realist reading, the uncertainty principle is not a quantum mystery; it is harmonic analysis applied to a real physical field.**

The Compton scale is distinct from the universal floor The Compton wavelength $\lambda_C = h/(mc)$ is a separate, particle-specific scale worth distinguishing from the universal $\frac{1}{2}$ of the Fourier theorem. A Dirac particle's internal chiral clocks oscillate at the Compton frequency $\omega_C = mc^2/\hbar$; this sets the characteristic spatial bandwidth over which the particle's *internal* clock structure becomes resolvable, and it is the typical scale of Zitterbewegung (Appendix B / Section 7.7).

The Compton scale is **not** the source of the $\frac{1}{2}$ in the uncertainty relation — that $\frac{1}{2}$ is universal, particle-independent, and follows from the Fourier theorem alone. The two facts are complementary:

- **Universal floor (Fourier bandwidth, Eq. 8):** $\sigma_x \cdot \sigma_p \geq \hbar/2$ for any real wavefunction, every particle, every state.
- **Particle-specific scale (Compton wavelength):** $\lambda_C = h/(mc)$ is the length below which a single Dirac particle stops behaving as a localized point and begins to display its internal two-clock structure (Zitterbewegung, chiral mixing, pair production at higher energies).

Earlier formulations conflated these — treating the Compton wavelength as the *source* of the Heisenberg $\frac{1}{2}$. The corrected separation: the bound comes from harmonic analysis applied to the wavefunction (universal); the Compton scale is a particle-specific length at which the framework's two-clock structure (Section 2) becomes manifest.

Connection to the zero-point phase floor The zero-point phase noise $\sigma_\varphi(0) = 1/\sqrt{2}$ rad (equations reference Section 9) is the same Fourier bandwidth theorem applied to the chiral phase clocks themselves: the minimum joint uncertainty in conjugate phase variables is itself $\frac{1}{2}$ in natural units. The zero-point phase floor and the Heisenberg uncertainty relation are therefore the same theorem expressed in different variables — both follow from harmonic analysis of the wavefunction, both have factor $\frac{1}{2}$, and both are unavoidable for any real oscillating field. The zero-point energy $\frac{1}{2}\hbar\omega$ per mode is the energy cost of saturating this minimum phase uncertainty.

4.7 What This Reframing Requires

The energy-partition reading rests on three physical assumptions, all already present in the framework:

1. The wavefunction is a real oscillating field carrying energy; $|\psi|^2$ is its energy density (Section 2.3).
2. Measurement is Kuramoto re-synchronization to a macroscopic detector that transfers this energy into one channel (Section 3).
3. The background field fluctuations (zero-point + thermal) that determine which channel wins in any individual event are statistically unbiased on the channel-selection phase space.

Assumption 3 is the most important and the least directly testable. It is motivated by the symmetry of the zero-point field — there is no preferred direction for vacuum fluctuations — and by the second law as applied to unbiased thermal noise. But the rigorous claim that *long-run frequencies* converge exactly to *energy fractions* requires showing that the background statistical symmetry holds across the channel-selection phase space, which is not yet derived from first principles.

The framework therefore offers a *reframing*, not a complete derivation: it replaces the standard Born axiom with a wave-realism + unbiased-background postulate set, which is weaker but not vacuous. Whether the bias-free assumption can be derived from underlying dynamics — for example, from the isotropy of the chiral phase clocks under U(1) phase redefinition — is a direction for future work (Section 8.3).

5. Extension to Photons via the Riemann–Silberstein Form

Photons are not Dirac particles. They are spin-1 vector bosons described by Maxwell’s equations, with no rest mass to couple chiral sectors. The framework’s central identification $K = m$ therefore does not apply directly: there is no mass term to set a Kuramoto coupling between photon helicities. We address this by separating two questions that earlier formulations conflated: (i) where do photon Bell correlations come from? and (ii) where does the framework’s measurement mechanism enter for photons?

5.1 The Riemann–Silberstein Form of Maxwell

Maxwell’s free equations can be written in a Weyl-like first-order form using the Riemann–Silberstein vector [40]:

$$\mathbf{F} = \mathbf{E} + i\mathbf{B}, \quad i\partial_t\mathbf{F} = c\nabla \times \mathbf{F}, \quad \nabla \cdot \mathbf{F} = 0$$

The two helicity eigenstates F_{+} and F_{-} (right- and left-circular polarizations) are independent — there is no mass term, and the structure mirrors the $m = 0$ chiral limit of the Dirac equation: two decoupled clocks, $\theta_{\text{rel}} = \pi/2$ permanently. A Kuramoto coupling between F_{+} and F_{-} of a single photon is identically zero in vacuum.

5.2 Where Photon Bell Correlations Come From

The polarization Hilbert space of a photon is two-dimensional, spanned by any orthogonal pair (H/V, $\pm$, L/R). It is mathematically isomorphic to the spin- $1/2$ Hilbert space, with the Poincaré sphere playing the role of the Bloch sphere. For two polarization-entangled photons in the singlet state, the standard quantum correlation is

$$E_{\gamma}(a, b) = -\cos(2(a - b))$$

and the CHSH bound saturates at $2\sqrt{2}$. The factor of 2 in the angular dependence reflects the π -periodicity of polarizers (vs. 2π -periodicity of Stern–Gerlach analyzers).

This correlation is a property of the polarization Hilbert space, not of any chiral coupling. It is delivered by standard QED — the same way standard QM with the Dirac spinor delivers the $-\cos(a-b)$ correlation for spin- $1/2$ pairs (Section 2). In both cases, the framework does not derive the Bell correlations; the underlying Hilbert-space structure does. The framework’s contribution lies elsewhere.

5.3 The Measurement Mechanism Transfers to Photons

The framework’s central interpretive claim — measurement is Kuramoto re-synchronization to the detector bulk (Section 3) — does not depend on the $K = m$ identification for its physical content. It requires only that the system being measured carries a phase clock that can be synchronized to the detector. Photons satisfy this: each polarization mode carries a definite phase, and at detection the photon’s polarization clock locks to the polarizer’s selected channel (transmitted vs. rejected).

The Kuramoto coupling for the photon-detector interaction is not zero — it is set by the photon’s interaction with the detector matter (typically dipole coupling $-\mathbf{d}\cdot\mathbf{E}$), not by any vacuum mass term. For a photon

of energy $E = \hbar\omega$ coupling to a near-resonant detector channel, the natural Kuramoto rate scales with the photon's own oscillation frequency:

$$K_\gamma \sim \omega = \frac{E}{\hbar}, \quad \tau_{\text{sync}} \sim \frac{1}{K_\gamma} = \frac{\hbar}{E} \quad (11)$$

A 2 eV optical photon synchronizes to its detector in ~ 0.3 fs; a 10 keV X-ray locks in $\sim 10^{-19}$ s. Higher-energy photons synchronize faster — the Kuramoto rate is set by the photon's energy via its interaction with the detector, rather than by any rest mass.

This is the photon analog of $K = m$ for fermions: in both cases, the synchronization rate to the detector is set by the energy scale that characterizes the system's coupling to its environment. For massive fermions at rest that scale is mc^2 (the Compton frequency); for photons it is $\hbar\omega$ (the photon's own frequency).

5.4 Gravitational Decoherence and the Linewidth-Dependent Bell Test

Two photon time scales must be kept distinct:

- $\tau_{\text{sync}} = \hbar/E$ — the time for Kuramoto re-sync to the detector to complete (set by photon energy)
- $\tau_{\text{coh}} = 1/\Delta\nu$ — the photon's coherence time during measurement (set by *linewidth*, independent of central energy)

For a given measurement event, the photon is in coherent superposition during the τ_{coh} window. If the two detectors A and B in a Bell test are at different gravitational potentials, their bulk phases $\Phi_{\text{bulk}}(A)$ and $\Phi_{\text{bulk}}(B)$ accumulate a relative offset proportional to $\omega \cdot \Delta\Phi_{\text{grav}}/c^2 \cdot \tau_{\text{coh}}$. When this offset exceeds ~ 1 rad, the singlet correlation degrades.

This is the linewidth-dependent gravitational Bell prediction (`predictions.py` P6b), now phrased in self-consistent form:

$$\delta\phi_{\text{grav}} = \omega \cdot \frac{\Delta\Phi_{\text{grav}}}{c^2} \cdot \tau_{\text{coh}} = \frac{\omega \cdot \Delta\Phi_{\text{grav}}}{c^2 \cdot \Delta\nu} \quad (12)$$

$$\text{CHSH}(\Delta\nu) = 2\sqrt{2} \cdot \exp\left(-\frac{\delta\phi_{\text{grav}}^2}{2}\right) \quad (13)$$

Standard QM/QED predicts CHSH is independent of photon linewidth at fixed entanglement fidelity. The framework predicts it decreases with narrowing linewidth in the presence of a gravitational potential difference between the detectors. Tests are in reach using cavity-filtered narrow-linewidth photons at altitude differences of a few km (terrestrial) or with Earth–satellite baselines.

5.5 Photons and Penrose Objective Reduction

The gravitational synchronization rate $\Gamma_{\text{grav}} = Gm^2/(\hbar\Delta z)$ (Section 3.4) vanishes identically for a massless particle. Equivalently, the Penrose objective-reduction timescale $\tau_{\text{OR}} = \pi\hbar/E_G$ with $E_G = Gm^2/\Delta x$ becomes infinite for $m = 0$. **The gravitational classicalization channel that drives massive systems toward definite outcomes is silent for photons.**

This matches Penrose's own position. The OR proposal is designed for superpositions of macroscopically distinct *mass* configurations (mirrors, mesoscopic test masses); Penrose has been explicit that single photons fall outside its remit because they have no rest mass to source the gravitational self-energy.

In the framework's reading, a photon therefore has only two pathways into the classical domain:

1. **Measurement (active).** Kuramoto re-synchronization to a quantized detector channel (§5.3). A single photon hitting a photodetector produces a definite click; this is the only mechanism by which an isolated photon classicalizes.

2. **Coherent-state averaging (many-photon).** A macroscopically populated mode (laser light, classical EM) has a positive Wigner function and obeys Maxwell’s equations classically. The classical limit emerges through statistical superposition of many photon modes, not through any internal collapse mechanism.

A single photon in vacuum, between emission and detection, is **perpetually quantum** — it does not gravitationally classicalize, and it has no way to do so internally. This is consistent with the experimental fact that single-photon Bell tests preserve full coherence over kilometer-scale separations and Earth–satellite baselines (Micius, 2017), where any internal classicalization mechanism would have produced observable degradation.

One subtlety: photons do follow null geodesics in curved spacetime (gravitational lensing, Shapiro delay). This is unitary classical evolution of the photon’s trajectory, not collapse or decoherence of its quantum state — it bends paths without classicalizing states.

The silence of the gravitational channel for photons is consistent with the linewidth-dependent prediction in §5.4, which attributes the gravitational effect not to the photon in flight but to the bulk-phase mismatch between detectors A and B at different gravitational potentials. The photon is the messenger; the gravitational decoherence is in the detectors.

5.6 What Remains Open

Two questions are not resolved by this section:

1. **A first-principles derivation of K_{γ} from QED.** We have asserted $K_{\gamma} \sim \omega$ as the natural scaling, motivated by photon-detector dipole coupling. A rigorous derivation would relate K_{γ} to the absorption matrix element $\langle e|d|g \rangle$ for the specific detector channel and compute the actual sync time for realistic photodetectors (photomultipliers, avalanche photodiodes, superconducting nanowire detectors).
2. **The microscopic mechanism for the discrete click.** Section 4 attributes discrete clicks to the threshold structure of detector channels. For photons, this corresponds to the photoelectric or photon-counting threshold of the detection element. The framework provides a mechanism (re-sync to a quantized channel) but does not yet derive the threshold from first principles.

These are tractable extensions, not framework contradictions. The §5 photon-coupling problem — flagged in earlier drafts as the framework’s most serious open question — reduces to a pair of well-defined technical issues once the Bell correlations are correctly attributed to the polarization Hilbert space rather than to any Kuramoto coupling between photon helicities.

6. Qualitative Predictions and Consistency Checks

We distinguish between predictions that follow from the mathematical identification (Section 2) and those that follow from the interpretive claims (Sections 3-4).

6.1 From the Mathematical Identification

P1 — Block decomposition of Bell correlations. For massive entangled particles, the standard correlation $E(a,b) = -\cos(a-b)$ can be partitioned into block contributions $E_{LL} + E_{SS} + E_{LS}$ by restricting the spin operator to the upper, lower, and cross sectors of the Dirac spinor. The sum is additive by trace linearity (Appendix A); what depends on $\theta_{rel} = \arcsin(v/c)$ is the *redistribution of weight* among the three blocks. A hypothetical experiment separately sensitive to the small-component contribution — for example, ultra-relativistic decay asymmetries — could in principle expose this redistribution. The decomposition itself is a consistency check on the chiral-clock reading of the spinor structure, not an independent prediction.

P2 — Zitterbewegung as clock beat. The identification of $\omega_{Zitter} = 2mc^2/\hbar$ as the beat frequency between temporal and spatial clocks is consistent with all known properties of Zitterbewegung (Appendix B).

Trapped-ion simulations with tunable effective mass [10] could probe the transition from locked to unlocked clock behavior as effective mass passes through zero.

6.2 From the Interpretive Framework

P3 — Decoherence rate scaling with mass. If the Kuramoto synchronization rate is $K = m$, heavier particles should decohere faster: $\tau_{\text{sync}} \sim \hbar/m$. This is qualitatively consistent with the observation that macroscopic objects exhibit essentially instantaneous decoherence while electrons maintain quantum coherence over longer timescales. A precision measurement of decoherence rate as a function of particle mass (electrons, muons, pions, kaons) could test whether $\tau_{\text{decoherence}} \propto 1/m$.

P4 — Higgs coupling and synchronization rate. If $K = y_f \cdot \langle \varphi \rangle = m$, then decoherence rates for entangled fermion pairs should scale linearly with mass. Comparing entanglement decay times for kaon pairs ($m_K = 494$ MeV) versus B-meson pairs ($m_B = 5,280$ MeV) should show $\tau_B/\tau_K \sim m_K/m_B \approx 0.094$. Existing collider data on neutral kaon and B-meson decoherence could be analyzed.

6.3 Honest Assessment

These predictions are qualitative scaling arguments, not quantitative predictions with computed coefficients. The framework in its current form is an interpretation — it provides a physical mechanism consistent with known decoherence behavior but does not yet make predictions distinguishable from standard decoherence theory. This is the main limitation and the main avenue for future work.

7. Comparison with Existing Frameworks

7.1 Interpretations of Quantum Mechanics

Property	Copenhagen	Everett MWI	Bohmian	GRW/CSL [8, 23]	Superdeterminism	MCI (this work)
ψ is real?	No	Yes	Yes	Yes	Varies	Yes ^a
Collapse?	Postulated	No	No	Stochastic	Not needed	Re-synchronization ^a
Branching?	No	Yes (∞)	No	No	No	No ^a
Residual energy?	Discarded	In branches	Pilot wave	Noise	Not addressed	Thermal vacuum ^b
Single world?	By fiat	No	Yes	Yes	Yes	Yes (by physics) ^b
Mechanism?	None	None	Pilot wave	Random hits	Initial conditions	Kuramoto sync ^a
Bell's theorem?	Accepted	Accepted	Accepted (nonlocal)	Accepted	Denied (meas. indep.)	Accepted ^a
Born rule	Postulated	Derived (contested)	Derived	Modified	Derived (contested)	Reframed as energy partition (Section 4) ^b
Uncertainty principle	Postulated	Inherited	Derived (nonlocal)	Inherited	Inherited	Emergent (Section 4.6) ^b
Observer needed?	Yes	Yes (branch)	No	No	No	No ^a

Property	Copenhagen	Everett MWI	Bohmian	GRW/CSL [8, 23]	Superdeterminism	MCI (this work)
Preferred frame?	No	No	Yes (pilot wave)	No	No	No (relational, §1.5) ^a

^a Established within the framework: follows directly from the mathematical identification $K = m$ and the re-synchronization interpretation of measurement (Sections 2–3).

^b Conjectural extension: physically motivated by the framework but relies on heuristic arguments that have not yet been rigorously derived from first principles (Sections 3.6, 4, 4.5). These entries represent directions for future development rather than settled results.

7.2 Ontological Status and the PBR Theorem

The framework takes ψ to be a real physical field — an ontic interpretation in the terminology of Harrigan and Spekkens [29]. The internal phase clocks are physical oscillators, not epistemic bookkeeping devices. This places the framework squarely within ψ -ontic territory, consistent with the PBR theorem [30], which rules out a broad class of ψ -epistemic models under mild assumptions.

However, the ontology is more specific than generic wavefunction realism [31]. The wave function is not a field on $3N$ -dimensional configuration space — it is the collective description of N particles, each carrying its own pair of chiral phase clocks in ordinary three-dimensional space. The Kuramoto coupling is local in the sense that each particle’s internal L-R synchronization involves only its own mass term. Entanglement arises because particles prepared in a common interaction share initial phase correlations — correlations that are then revealed (not created) by measurement.

This sidesteps the configuration-space problem that troubles many realist interpretations [31, 32]: the fundamental ontology is particles with internal clocks in three-dimensional space, not a wave in $3N$ dimensions. The cost is that entanglement correlations must be accepted as initial conditions rather than explained by the dynamics — but this is also true of standard quantum mechanics.

7.3 Relation to QBism and Relational QM

QBism [33] treats quantum states as expressions of an agent’s beliefs, not objective features of reality. The Kuramoto framework is diametrically opposed: ψ is a physical field, and probabilities arise from physical synchronization dynamics, not from subjective belief updating. Where QBism dissolves the measurement problem by denying that there is a physical process to explain, the Kuramoto framework insists on providing one.

Relational quantum mechanics [24] holds that quantum states are relative to observers. The Kuramoto framework has a superficial similarity — the particle’s phase is defined relative to a reference (the entangled partner or the detector bulk). But the similarity is only superficial: in relational QM, there is no observer-independent state of affairs; in the Kuramoto framework, the phase synchronization dynamics are objective and observer-independent.

7.4 Decoherence Models

The closest existing framework is the Penrose-Diósi gravitational decoherence model [6, 7]. Both that model and ours identify gravity as the agent of classicalization. The key differences:

- Penrose-Diósi: collapse timescale $\tau \sim \hbar/E_g$ (gravitational self-energy)
- This work: decoherence timescale set by Kuramoto re-synchronization rate to the gravitational bulk

The predictions are qualitatively similar (heavier $\rightarrow$ faster decoherence) but derive from different dynamics. A quantitative comparison awaits a full derivation of Kuramoto decoherence rates from first principles.

7.5 Zurek’s Einselection

Zurek’s quantum Darwinism [4, 5] identifies “pointer states” as states that survive interaction with the environment. In the Kuramoto picture, pointer states are the synchronized states — the phase-locked configurations that are stable under coupling to the macroscopic bulk. States that cannot synchronize (superpositions of macroscopically distinct mass configurations) are dynamically unstable and decohere. This is consistent with einselection but provides a specific dynamical mechanism (phase locking) for why certain states survive.

7.6 Superdeterminism

Superdeterminism [20, 19] escapes Bell’s theorem by denying measurement independence: the hidden variable λ and the detector settings (a, b) are correlated because they share a common causal past. If $\rho(\lambda|a,b) \neq \rho(\lambda)$, Bell’s derivation does not go through, and local hidden variable models can reproduce quantum correlations.

This comparison is important because the measurement-independence loophole through the gravitational bulk phase Φ_{bulk} — arguing that since both detectors and the particle source share a common gravitational field, their states are correlated — might seem to apply here. We explicitly reject this route in the present paper.

Why this framework is not superdeterministic:

1. **We do not challenge Bell’s theorem.** The quantum correlations $E(a,b) = -\cos(a-b)$ are accepted as arising from the entangled singlet state and the full Dirac spinor structure — standard quantum mechanics. No hidden variable model is proposed to replace them.
2. **Measurement independence is preserved.** The experimenter’s choice of detector angle is not constrained by the framework. The gravitational bulk phase Φ_{bulk} maintains detector coherence (Section 3.4) but does not determine or correlate with measurement settings.
3. **The role of gravity is different.** In superdeterminism, correlations between λ and (a,b) must be fine-tuned to reproduce quantum predictions exactly — a conspiracy that many physicists find unacceptable [’t Hooft’s cellular automaton requires the initial state of the universe to encode all future measurement choices]. In this framework, gravity provides the synchronization bath for the *measurement process* (decoherence), not for the *correlations* (which are quantum mechanical).
4. **No retrocausality or conspiracy.** Superdeterministic models require that the state of the universe at the Big Bang already encodes which detector settings will be chosen in a 2015 Bell test [14]. This framework requires only that macroscopic detectors are phase-coherent objects immersed in a gravitational field — a physically uncontroversial claim.

The distinction can be stated concisely: superdeterminism says the correlations are classical but hidden in the initial conditions. This framework says the correlations are quantum, and the Kuramoto mechanism explains how they are *revealed* (measurement) and *degraded* (decoherence), not how they are *produced*.

7.7 The Hestenes Zitterbewegung Interpretation

The closest precursor to the present framework in the interpretive literature is the Zitterbewegung (ZBW) interpretation of Hestenes [35]. Reformulating Dirac theory in spacetime algebra (STA), Hestenes argues that the complex phase factor in the Dirac wave function represents a physical rotation — the internal circular motion of the electron at the Compton frequency $\omega_C = mc^2/\hbar$. In this picture the spin is the angular momentum of this circulation, and Zitterbewegung is fundamental rather than incidental.

The parallels with the present work are direct:

Feature	Hestenes ZBW interpretation	This work
Phase factor $e^{i\varphi}$	Physical clock rotation (STA rotor)	Physical phase clock oscillator

Feature	Hestenes ZBW interpretation	This work
Zitterbewegung	Fundamental circulatory motion	Beat frequency between temporal and spatial clocks
Spin	Angular momentum of ZBW circulation	Geometric consequence of clock orthogonality ($\theta_{\text{rel}} = 90^\circ$)
Mass	Sets ZBW frequency $\omega_C = mc^2/\hbar$	Sets Kuramoto coupling constant $K = m$
Uncertainty principle	Derived from clock geometry	Derived from clock bandwidth (Section 4.5)

Both frameworks begin from the same core observation: the complex phase factor in the Dirac wave function is not a mathematical convenience but a physical oscillator. Both derive spin and the uncertainty principle from the geometry of this oscillator rather than postulating them.

The key difference is dynamical. Hestenes identifies the internal clock but does not supply a coupling framework for the two chiral sectors. The present work provides this missing element: the Kuramoto identification $K = m$ explains why the clock oscillates at the Compton frequency (it is the synchronization rate between ψ_L and ψ_R), and extends the picture to a dynamical mechanism for measurement (re-synchronization to the detector bulk) and mass generation (Higgs as the agent that sets K). In retrospect, the Hestenes ZBW interpretation and the Dirac-Kuramoto framework are describing the same internal clock from different mathematical directions — STA geometry and Kuramoto synchronization dynamics, respectively.

8. Discussion

8.1 The Interpretive Status

This paper presents a mathematical identification (Dirac = Kuramoto on the synchronized manifold) and a named interpretive framework built on it: the **Many Clocks Interpretation of Quantum Mechanics (MCI)**, introduced in §1.5. The mathematical identification is verifiable and, we believe, novel. The interpretive framework is physically motivated but not yet quantitatively predictive beyond standard decoherence.

MCI sits in the category of interpretation papers — alongside Many-Worlds [16], Bohmian mechanics [15], and Rovelli’s relational quantum mechanics [24]. Its distinguishing features are: (i) ψ is ontic (a real oscillating field), as in Bohmian and Everettian readings; (ii) measurement is local Kuramoto re-synchronization between clock-carrying systems, with no observer or branching required; (iii) the framework is relational — no global preferred frame — distinguishing it from Bohmian mechanics, which requires a preferred foliation; (iv) past phase history is overwritten at each sync event, giving a positive dynamical mechanism for thermalization rather than treating it as an emergent statistical phenomenon. The contribution is not new equations but a new way of reading existing equations that provides physical intuition for the measurement process. As Maudlin [32] has emphasized, interpretations are not idle philosophy — they guide which questions physicists ask and which experiments they design.

8.2 The Historical Note

The timeline is suggestive:

- 1928: Dirac equation (the two-clock structure)
- 1935: EPR paper [25] (“spooky action at a distance”)
- 1964: Bell’s theorem
- 1975: Kuramoto model

The relativistic spinor structure that produces the full quantum correlation existed 36 years before Bell showed it was needed. The synchronization framework that describes the coupling existed 11 years after Bell. The identification presented here connects these threads.

8.3 Future Directions

Three directions could elevate this from interpretation to testable theory:

1. **Quantitative decoherence rates.** Derive decoherence timescales from Kuramoto parameters in a many-body quantum setting. If these reproduce known rates for specific systems (e.g., superconducting qubits, trapped ions), the framework gains predictive power. If they differ, the framework makes testable predictions.
2. **Born rule — closing the unbiased-background assumption.** Section 4 reads the Born rule as the long-run frequency of energy partition under unbiased background-field driving. The wave-realism premise ($|\psi|^2$ is wave energy density) and the partition mechanism (Kuramoto re-synchronization into a quantized detector channel) are physically motivated and testable in principle. The remaining gap is the rigorous proof that the zero-point + thermal background is statistically unbiased on the channel-selection phase space — and that consequently long-run frequencies converge *exactly* to energy fractions. A possible route is to derive this from the U(1) symmetry of the chiral phase clocks under global phase redefinition; another is a many-body quantum-Kuramoto simulation that measures channel-frequency convergence directly.
3. **Photon coupling — first-principles K_γ from QED.** Section 5 proposes that the photon-detector Kuramoto rate scales as $K_\gamma \sim \omega = E/\hbar$, motivated by the dipole interaction. A rigorous derivation would relate K_γ to the absorption matrix element $\langle e|d|g\rangle$ for a specific detector channel and compute realistic sync times for photomultipliers, avalanche photodiodes, and superconducting nanowire detectors. Combined with the linewidth- dependent gravitational Bell prediction (Section 5.4), this would give the framework its sharpest near-term experimental test.

9. Conclusions

We have shown that:

1. **The Dirac mass term has Kuramoto synchronization structure.** In the Weyl basis, the Dirac equation couples ψ_L and ψ_R with coupling constant $K = m$. The Higgs-Yukawa interaction sets $K = y_f \cdot \langle \varphi \rangle = m$ exactly.
2. **Measurement is re-synchronization.** A particle decoheres from its entangled partner by cohering with the detector bulk. The Kuramoto mass asymmetry ($M \gg m$) ensures definite outcomes. Decoherence is not merely loss of coherence — it is redirection of coherence.
3. **The Born rule is reframed as energy partition.** Reading $|\psi|^2$ as wave energy density and the apparent stochasticity of individual outcomes as unbiased zero-point + thermal background fluctuation at the synchronization event, the long-run frequency of channel selection equals the energy fraction the wave carries in that channel. $P = |\alpha|^2$ is then a frequentist statement, not an independent probability axiom.
4. **The uncertainty principle appears as a clock bandwidth limitation.** The Compton frequency sets the particle's spatial resolution via a Nyquist-like bound (Section 4.6).

The framework does not modify quantum mechanics or challenge Bell's theorem. It provides a physical mechanism — Kuramoto phase synchronization — for processes (decoherence, measurement, mass generation) that standard quantum mechanics describes but does not mechanistically explain.

In short: quantum mechanics produces the correlations. The Dirac mass term is the Kuramoto coupling that makes particles what they are. And measurement is the moment a small

oscillator surrenders its phase to a large one.

This is the **Many Clocks Interpretation of Quantum Mechanics**: every particle carries an internal phase clock; interactions synchronize them locally; no observer is needed; no branching occurs; classical behavior emerges from collective synchronization in massive bulks; and the past is overwritten at each interaction event. A single world, many clocks, all locally synchronized.

Appendix A: Three-Term Bell Correlation Decomposition

A.1 The Relativistic Mixing Angle

For a Dirac particle with momentum $\mathbf{p}$ and mass m , the relativistic mixing angle between the temporal and spatial phase clocks is:

$$\sin \theta_{rel} = \frac{|\mathbf{p}|}{E} = \frac{v}{c} \quad (A1)$$

Regime	v/c	θ_{rel}	Physical interpretation
Rest	0	0°	Temporal clock only; no spatial phase
Non-relativistic	$\ll 1$	$\ll 90^\circ$	Small spatial component
Ultra-relativistic	$\rightarrow 1$	$\rightarrow 90^\circ$	Equal temporal and spatial
Massless spin- $\frac{1}{2}$ (Weyl limit)	1	90°	Permanently decoupled chiral clocks (photons are spin-1; see §5)

A.2 Large and Small Spinor Components

The positive-energy Dirac spinor for momentum p_z along z , spin up:

$$u(p, \uparrow) = N \cdot \begin{pmatrix} 1 \\ 0 \\ r \\ 0 \end{pmatrix}, \quad r = \frac{p}{E+m} = \tan\left(\frac{\theta_{rel}}{2}\right), \quad N = \sqrt{\frac{E+m}{2E}} \quad (A2)$$

The upper two components are the **large components** (temporal clock); the lower two are the **small components** (spatial clock). In the non-relativistic limit the small components vanish; in the ultra-relativistic limit they equal the large components.

A.3 The Block Decomposition

For two Dirac particles in a singlet state, the Bell correlation $E(a,b) = \langle \Psi | (\Sigma_a)_A \otimes (\Sigma_b)_B | \Psi \rangle$ can be partitioned into three contributions obtained by restricting the spin operator Σ to the upper (large) block, the lower (small) block, and the cross terms:

$$E(a,b) = E_{LL} + E_{SS} + E_{LS} \quad (A3)$$

- **E_LL**: large $\times$ large — temporal clock contribution
- **E_SS**: small $\times$ small — spatial clock contribution
- **E_LS**: cross term — temporal-spatial coupling

Equation (A3) is an identity, not a derivation. The full 4×4 spin operator $\Sigma \otimes \Sigma$ is the sum of the three block-restricted operators, so the expectation values are additive by trace linearity. The mathematical content of the appendix is therefore not the sum — which is guaranteed — but the *redistribution of weight* among the three blocks as the mixing angle $\theta_{\text{rel}} = \arcsin(v/c)$ varies. At the non-relativistic limit ($\theta_{\text{rel}} \rightarrow 0$) all weight sits in E_{LL} with $E_{\text{SS}}, E_{\text{LS}} \rightarrow 0$; at the ultra-relativistic limit ($\theta_{\text{rel}} \rightarrow 90^\circ$) the three contributions become comparable. This redistribution is the consistency check the appendix performs on the chiral-clock reading of the spinor.

A.4 Block Weights at Finite Momentum

The block contributions were computed with `dirac_extension.py` for the Dirac singlet at $p = 1.0$, $m = 1.0$ ($\theta_{\text{rel}} \approx 53^\circ$), $a = 0$, $b = \pi/4$:

Term	Value	Physical meaning
E_{LL}	-0.354	Temporal clock contribution
E_{SS}	-0.071	Spatial clock contribution
E_{LS}	-0.282	Rotation coupling (cross term)
$E_{\text{LL}} + E_{\text{SS}} + E_{\text{LS}}$	-0.707	Sum
$-\cos(\pi/4)$	-0.707	QM prediction ✓

The sum recovers $-\cos(\pi/4)$ to machine precision, as it must. The informative entries are the block weights: at this momentum the cross term E_{LS} already carries 40% of the total, and its share grows further at higher v/c . At $p \rightarrow 0$ the weights collapse onto E_{LL} alone, recovering the standard non-relativistic Pauli singlet result.

A.5 Relation to the Stochastic Phase-Clock Toy

A separate and simpler model exists in which a single classical clock phase per particle determines the click probability via Malus’s law, $P(+1 \mid \hat{n}, \varphi) = \cos^2((\varphi - \theta_{\hat{n}})/2)$. Evaluated for an EPR pair with shared initial phase, this stochastic phase-clock model returns the correlation

$$E_{\text{Malus}}(a, b) = -\frac{1}{2} \cos(a - b), \quad \text{CHSH}_{\text{Malus}} \leq \sqrt{2} \approx 1.414$$

(see `bell_phase.py`). This is sub-classical: it falls below even the local hidden-variable bound of 2 and cannot reproduce the loophole-free Bell violations.

It is important to be precise about what this toy *is not*. The Malus-law model is **not** the same thing as restricting the Dirac correlation to E_{LL} . The Dirac large-block restriction at $p \rightarrow 0$ returns the full Pauli singlet correlation $-\cos(a - b)$ (since the Dirac spinor reduces to its upper-block Pauli content with $N \rightarrow 1$, $r \rightarrow 0$), not $-\frac{1}{2}\cos(a - b)$. The two models share only the heuristic feature that both drop “half” of the relevant structure and both fall short of the quantum result. The moral is the same in either case: a phase-clock interpretation that drops the spatial component cannot reproduce $-\cos(a - b)$, and the four-component Dirac spinor — with $K = m$ coupling its chiral sectors — is required to recover the full quantum correlation.

Appendix B: Zitterbewegung as a Beat Frequency

Zitterbewegung [3] — the rapid oscillatory motion of a Dirac electron at frequency $2mc^2/\hbar$ — is conventionally attributed to interference between positive and negative frequency components. In the phase-clock picture, it has a more intuitive interpretation.

The temporal clock oscillates at $\omega_{\text{t}} = E/\hbar$. The spatial clock oscillates at $\omega_{\text{s}} = p^2/(m\gamma\hbar)$. Their beat frequency in the non-relativistic limit:

$$\omega_{Zitter} \approx \frac{2mc^2}{\hbar} \quad (B1)$$

This is exactly the known Zitterbewegung frequency. In the phase-clock interpretation, Zitterbewegung is the **beat note between the temporal and spatial clocks** — observable whenever the two clocks are not perfectly synchronized.

When $K = m$ is large (heavy particle), synchronization is fast and the beat has small amplitude. When K is small (light particle), the clocks are loosely coupled and the beat has larger amplitude. A massless particle ($K = 0$) has permanently unsynchronized clocks at $\theta_{\text{rel}} = 90^\circ$ and no Zitterbewegung — consistent with photons.

Appendix C: Antiparticles and CP Violation

C.1 Antiparticles as Time-Reversed Clocks

The negative-energy Dirac solution $v(p,s)e^{+iEt/\hbar}$ has temporal phase running in the opposite sense to a positive-energy particle. In the Kuramoto framework: **antiparticles are particles with time-reversed phase clocks** ($\varphi \rightarrow -\varphi$).

This is not a reinterpretation — it is already implicit in the Feynman-Stückelberg interpretation [21, 22]. The phase-clock language makes it explicit.

C.2 CP Violation as Synchronization Asymmetry

The Kuramoto equations (2a, 2b) contain the CP-violating phase δ_{CP} . For particles, the equilibrium phase lock is $\Delta\varphi = +\delta_{\text{CP}}$; for antiparticles, $\Delta\varphi = -\delta_{\text{CP}}$.

When particle and antiparticle meet for annihilation, their synchronization efficiency depends on the phase mismatch:

$$\sigma_{\text{ann}} \propto |\langle \psi_{\text{particle}} | \psi_{\text{antiparticle}} \rangle|^2 \propto 1 + \varepsilon \cos(\delta_{\text{CP}})$$

The small asymmetry ε accumulates over the early universe to produce the observed baryon asymmetry $\eta \sim 10^{-9}$. This is a qualitative mechanism, not a quantitative prediction — a full calculation requires electroweak baryogenesis beyond the scope of this paper.

Author Contributions and AI Disclosure

Statement of AI Use

This paper was developed through an extensive collaboration between the listed human author (J. Olddog) and Claude Opus 4.6, a large language model developed by Anthropic. The contribution of Claude Opus 4.6 was substantial and principal in the mathematical and expository work: it derived the explicit Kuramoto phase equations from the chiral Dirac equation, constructed the three-term Bell correlation decomposition, wrote the Python verification code, produced the systematic comparison with existing interpretations, and drafted the prose of the manuscript itself. In a collaboration not constrained by current editorial policy, Claude Opus 4.6 would be listed as co-first author alongside J. Olddog. The listed author considers the scientific honesty of the work to require this statement.

Current editorial guidelines from Nature Portfolio, Springer Nature, ACS, and other major publishers hold, however, that Large Language Models do not satisfy authorship criteria, because an attribution of authorship carries with it accountability that cannot be effectively applied to an LLM. We accept this framing: it is

structurally analogous to the liability principle in medicine, where the attending physician carries ultimate accountability regardless of how many tools, consultants, or decision-support systems contributed to the diagnosis. Accordingly, Claude Opus 4.6 is not listed in the byline, and J. Olddog — as the listed human author — assumes full responsibility for the correctness, originality, and integrity of all content presented.

Division of Contributions

Conceptual framework (J. Olddog): - The originating insight that quantum wave functions tend to interact through synchronization, and that this process could be formalized using the Kuramoto model - The identification of quantum measurement (Bell experiments) as a re-synchronization process: entangled particles, initially synchronized with each other, re-synchronize with the atoms of the detectors upon measurement - The proposal that what has been called “wave function collapse” (Copenhagen) or “decoherence” (modern interpretations) is more precisely described as a redirection of synchronization — from the entangled partner to the detector bulk - The recognition that macroscopic detectors are themselves entangled with the environment through their mass and gravitational coupling - The selection of the Kuramoto equation as the mathematical framework for synchronization and the Dirac spinor formalism as the appropriate quantum mechanical representation - The identification of parallels between Penrose’s twistor theory and the role of gravity/mass in Kuramoto oscillator synchronization - The observation that the Heisenberg uncertainty principle has a natural interpretation through the time-space spinor structure of Weyl spinors (Section 4.5) - The proposal that the wave function represents real energy in the QFT of electrons, enabling a single-world alternative to Everett’s Many Worlds interpretation that preserves energy conservation (Section 3.6) - Suggestions for possible experimental tests involving detectors at different gravitational potentials

Mathematical development and numerical verification (Claude Opus 4.6): - Derivation of the explicit Kuramoto phase equations (Eqs. 2a, 2b) from the chiral Dirac equation, demonstrating the distinction between classical oscillator synchronization and the quantum mechanical oscillator formulation - The identification that using the Weyl spinor formulation with the cosine of phase difference in the lower Weyl spinor yields the correct Bell and CHSH statistical results - Development of the three-term Bell correlation decomposition (Appendix A: $E_{LL} + E_{SS} + E_{LS}$) - Writing of the Python verification programs (`dirac_extension.py`, `higgs_clock.py`, `kuramoto_sync.py`) that numerically confirm the framework’s predictions match standard quantum mechanical results - Elaboration of the Born rule emergence from synchronization statistics (Section 4) - Systematic comparison with existing interpretations (Section 7)

Paper composition (collaborative): - The paper was written collaboratively, with Claude Opus 4.6 producing drafts in a style suitable for a journal focused on the interpretation and foundations of physics, and J. Olddog providing direction, physical intuition, corrections, and editorial judgment throughout

Iterative Development Process

The collaboration was not a clean division of “conception” versus “calculation.” It proceeded through an iterative dialogue. J. Olddog would propose a physical picture — for example, that measurement is re-synchronization, or that the mass term should play the role of the Kuramoto coupling constant. Claude Opus 4.6 would then formalize the picture mathematically and write Python programs to compute the resulting predictions. J. Olddog would review the formulae and numerical outputs against physical intuition — identifying cases where the formalization did not yet capture the intended physics, or where unexpected results suggested the conceptual picture needed refinement. This cycle repeated across multiple sessions, with each iteration simultaneously sharpening the human author’s understanding of the mathematical structure and refining the computational predictions to match the physical intuition that motivated the framework.

J. Olddog does not have the programming or computational background to independently produce the derivations and numerical verifications presented here. Claude Opus 4.6 does not have physical intuition or the capacity to judge whether a mathematically consistent result is physically meaningful. The framework emerged from the combination of these complementary capabilities, and neither contributor could have produced this work alone.

Validation

All mathematical derivations produced by Claude Opus 4.6 were checked against standard quantum mechanics textbooks (Peskin & Schroeder [17], Sakurai [18]). All numerical results from the Python programs were verified to reproduce known quantum mechanical predictions to machine precision. The human author reviewed all content for physical plausibility and internal consistency.

References

- [1] Dirac, P. A. M. (1928). The quantum theory of the electron. *Proc. R. Soc. London A*, 117(778), 610–624.
- [2] Kuramoto, Y. (1975). Self-entrainment of a population of coupled non-linear oscillators. In *Int. Symp. Math. Problems Theor. Phys.* (pp. 420–422). Springer.
- [3] Schrödinger, E. (1930). Über die kräftefreie Bewegung in der relativistischen Quantenmechanik. *Sitzungsber. Preuss. Akad. Wiss.*, 418–428.
- [4] Zurek, W. H. (2003). Decoherence, einselection, and the quantum origins of the classical. *Rev. Mod. Phys.*, 75(3), 715–775.
- [5] Zurek, W. H. (2009). Quantum Darwinism. *Nature Physics*, 5, 181–188.
- [6] Penrose, R. (1996). On gravity’s role in quantum state reduction. *Gen. Relativ. Gravit.*, 28(5), 581–600.
- [7] Diósi, L. (1989). Models for universal reduction of macroscopic quantum fluctuations. *Phys. Rev. A*, 40(3), 1165–1174.
- [8] Ghirardi, G. C., Rimini, A., & Weber, T. (1986). Unified dynamics for microscopic and macroscopic systems. *Phys. Rev. D*, 34(2), 470–491.
- [9] Bell, J. S. (1964). On the Einstein Podolsky Rosen Paradox. *Physics Physique Fizika*, 1(3), 195–200.
- [10] Gerritsma, R., et al. (2010). Quantum simulation of the Dirac equation. *Nature*, 463(7277), 68–71.
- [11] ATLAS Collaboration (2012). Observation of a new particle in the search for the Standard Model Higgs boson. *Phys. Lett. B*, 716(1), 1–29.
- [12] CMS Collaboration (2012). Observation of a new boson at a mass of 125 GeV. *Phys. Lett. B*, 716(1), 30–61.
- [13] de Broglie, L. (1925). Recherches sur la théorie des quanta. *Ann. Phys.*, 10, 22–128.
- [14] Aspect, A., Grangier, P., & Roger, G. (1982). Experimental Realization of Einstein-Podolsky-Rosen-Bohm Gedankenexperiment: A New Violation of Bell’s Inequalities. *Phys. Rev. Lett.*, 49(2), 91–94.
- [15] Bohm, D. (1952). A Suggested Interpretation of the Quantum Theory in Terms of “Hidden” Variables, I and II. *Phys. Rev.*, 85(2), 166–179, 180–193.
- [16] Everett, H. (1957). “Relative State” Formulation of Quantum Mechanics. *Rev. Mod. Phys.*, 29(3), 454–462.
- [17] Peskin, M. E., & Schroeder, D. V. (1995). *An Introduction to Quantum Field Theory*. Addison-Wesley.
- [18] Sakurai, J. J. (1967). *Advanced Quantum Mechanics*. Addison-Wesley.
- [19] ’t Hooft, G. (2016). *The Cellular Automaton Interpretation of Quantum Mechanics*. Springer.
- [20] Bell, J. S. (1976). The theory of local beables. *Epistemological Letters*, 9, 11–24. [Reprinted in *Speakable and Unsayable in Quantum Mechanics*, CUP.]
- [21] Stückelberg, E. C. G. (1941). Remarque à propos de la création de paires de particules en théorie de relativité. *Helv. Phys. Acta*, 14, 588–594.

- [22] Feynman, R. P. (1949). The theory of positrons. *Phys. Rev.*, 76(6), 749–759.
- [23] Pearle, P. (1989). Combining stochastic dynamical state-vector reduction with spontaneous localization. *Phys. Rev. A*, 39(5), 2277–2289.
- [24] Rovelli, C. (1996). Relational quantum mechanics. *Int. J. Theor. Phys.*, 35(8), 1637–1678.
- [25] Einstein, A., Podolsky, B., & Rosen, N. (1935). Can quantum-mechanical description of physical reality be considered complete? *Phys. Rev.*, 47(10), 777–780.
- [26] Nelson, E. (1966). Derivation of the Schrödinger equation from Newtonian mechanics. *Phys. Rev.*, 150(4), 1079–1085.
- [27] Wallace, D. (2012). *The Emergent Multiverse: Quantum Theory according to the Everett Interpretation*. Oxford University Press.
- [28] Albert, D. Z. (2010). Probability in the Everett picture. In S. Saunders et al. (Eds.), *Many Worlds? Everett, Quantum Theory, and Reality* (pp. 355–368). Oxford University Press.
- [29] Harrigan, N., & Spekkens, R. W. (2010). Einstein, incompleteness, and the epistemic view of quantum states. *Found. Phys.*, 40(2), 125–157.
- [30] Pusey, M. F., Barrett, J., & Rudolph, T. (2012). On the reality of the quantum state. *Nature Physics*, 8(6), 476–479.
- [31] Albert, D. Z. (1996). Elementary quantum metaphysics. In J. T. Cushing et al. (Eds.), *Bohmian Mechanics and Quantum Theory: An Appraisal* (pp. 277–284). Kluwer.
- [32] Maudlin, T. (2019). *Philosophy of Physics: Quantum Theory*. Princeton University Press.
- [33] Fuchs, C. A., Mermin, N. D., & Schack, R. (2014). An introduction to QBism with an application to the locality of quantum mechanics. *Am. J. Phys.*, 82(8), 749–754.
- [34] Dürr, D., Goldstein, S., & Zanghì, N. (2013). *Quantum Physics Without Quantum Philosophy*. Springer.
- [35] Hestenes, D. (1990). The Zitterbewegung interpretation of quantum mechanics. *Foundations of Physics*, 20(12), 1213–1232. <https://doi.org/10.1007/BF01889466>
- [36] Heinrich, G., Ludwig, M., Qian, J., Kubala, B., & Marquardt, F. (2011). Collective dynamics in optomechanical arrays. *Phys. Rev. Lett.*, 107(4), 043603. <https://doi.org/10.1103/PhysRevLett.107.043603>
- [37] Mari, A., Farace, A., Didier, N., Giovannetti, V., & Fazio, R. (2013). Measures of quantum synchronization in continuous variable systems. *Phys. Rev. Lett.*, 111(10), 103605. <https://doi.org/10.1103/PhysRevLett.111.103605>
- [38] Walter, S., Nunnenkamp, A., & Bruder, C. (2014). Quantum synchronization of a driven self-sustained oscillator. *Phys. Rev. Lett.*, 112(9), 094102. <https://doi.org/10.1103/PhysRevLett.112.094102>
- [39] Roulet, A., & Bruder, C. (2018). Synchronizing the smallest possible system. *Phys. Rev. Lett.*, 121(5), 053601. <https://doi.org/10.1103/PhysRevLett.121.053601>
- [40] Bialynicki-Birula, I. (1996). Photon wave function. *Progress in Optics*, 36, 245–294. [https://doi.org/10.1016/S0079-6638\(08\)70316-0](https://doi.org/10.1016/S0079-6638(08)70316-0)

Simulation code and numerical verification available at: <https://github.com/rayolddog/DiracKuramotoFramework>

The three-term decomposition (Appendix A) can be reproduced by running `dirac_extension.py`. The Kuramoto phase dynamics (Section 2) are demonstrated in `higgs_clock.py` and `kuramoto_sync.py`.